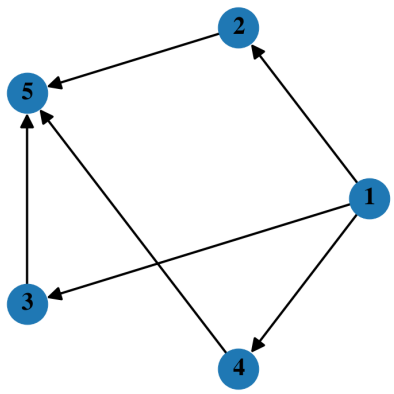
You should attempt ALL questions. Marks available are shown next to the questions.

In completing this assessment:

- **You may use books and notes.**
- **You may use calculators and computers, but you must show your working for any calculations you do.**
- **You may use the Internet as a resource, but not to ask for the solution to an exam question or to copy any solution you find.**
- **You must not seek or obtain help from anyone else.**

Question 1 [40 marks].
Consider the directed network G, with $N = 5$ nodes and $L = 6$ links, shown in figure.



a) Write the adjacency matrix $\mathbf{A}$ of the network (*Use the convention that entry a_{ij} is equal to 1 if node j points to node i*). Is matrix $\mathbf{A}$ symmetric? **[4]**

b) Determine the in-degree sequence $\{k_1^{in}, k_2^{in}, k_3^{in}, k_4^{in}, k_5^{in}\}$ and the out-degree sequence $\{k_1^{out}, k_2^{out}, k_3^{out}, k_4^{out}, k_5^{out}\}$ of the network. Find the nodes with the largest in-degree and the ones with the largest out-degree. **[6]**

c) Determine the in-degree distribution $P^{in}(k)$ and the out-degree distribution $P^{out}(k)$. **[4]**

d) State the definition of the Katz centrality x_i of a node i of a network. Find the Katz-centrality for all the nodes of the network G. **[7]**

e) How many weakly and how many strongly connected components are there in the network G? Which are the nodes belonging to each one of these components? **[3]**

f) Consider now the underlying undirected network G^u of G, i.e. the network obtained from G by neglecting the directions of the links. Write the adjacency matrix $\mathbf{A}^u$ of G^u. How many links has network G^u? **[4]**

g) Write the $N \times N$ distance matrix $\mathbf{d}$ of network G^u (*Remember that entry d_{ij} indicates the length of the shortest paths between node i and j*). What it the diameter D of G^u? **[4]**

h) Determine the degree distribution $P(k)$ of network G^u. **[2]**

i) State the definition of the betweenness centrality b_i of a node i of an undirected network. Evaluate the betweenness centrality b_1 of node 1 and the betweenness centrality b_4 of node 4 in network G^u. **[6]**

Question 2 [35 marks].
Let $n_0 = 10$, $m = 3$ and $a > 0$, and consider the following model to grow simple networks.
At time $t = 1$ we start with a complete network with n_0 nodes.
At each time step $t > 1$ a new node is added to the network. The node arrives together with m new links, which are connected to m different nodes already present in the network. The probability Π_i that a new link is connected to node i is:

$$\Pi_i = \frac{k_i + a}{Z} \qquad \text{with} \quad Z = \sum_{j=1}^{N(t-1)} (k_j + a)$$

where k_i is the degree of node i, and $N(t-1)$ is the number of nodes in the network at time $t-1$.

(a) Find an expression for the number of nodes, $N(t)$, and the number of links, $L(t)$, in the network as a function of time t. **[4]**

(b) What is the average node degree $\langle k \rangle$ at time t? What is the average node degree in the limit $t \to \infty$? **[4]**

(c) Find an expression for the value of Z as a function of time t. **[3]**

(d) Write down the differential equation governing the time evolution of the degree k_i of node i for $t \gg 1$ in the mean-field approximation. Solve this equation with the initial condition $k_i(t_i) = m$, where t_i is the time of arrival of node i. **[7]**

(e) Derive the degree distribution $P(k)$ of the network for $t \gg 1$ in the mean-field approximation. **[7]**

(f) Does the model produce scale-free networks? If so, what is the value of the degree exponent γ? **[4]**

(g) Write down the master equation of the model, i.e. the equation that describes the evolution of the average number $N_k(t)$ of nodes that at time t have degree k. **[6]**

End of Paper.